

MARMARA UNIVERSITY

FACULTY OF ENGINEERING



EE4077

FUNDAMENTALS OF MACHINE LEARNING

HOMEWORK I
SOLUTIONS

SOLUTION FOR QUESTION 1

a)

Derivative of function $f(x,z) = xcos(z)e^{-3x+z}$ can be computed as follows

$$f(x,z) = xcos(z)e^{-3x+z}$$

$$\frac{d}{dx}f(x,z) = \frac{d}{dx}xcos(z)e^{-3x+z} = \cos(z)\frac{d}{dx}(xe^{-3x+z}) = \cos(z)(e^{-3x+z} + x(-3)e^{-3x+z})$$

$$\frac{d}{dx}f(x,z) = \cos(z)(e^{-3x+z}(1-3x))$$

b)

From the Properties of variance following conditions are satisfied

$$Var(cX) = c^2Var(X)$$

$$Var(c+X) = Var(x)$$

Therefore,

$$Var(2X) = 4Var(X) = 4\sigma^2$$

$$Var(c+X) = Var(x) = \sigma$$

c)

According to joint distribution, conditional probability can be calculated as follows

$$P(X = T \mid Y = b) = \frac{0.1}{0.1 + 0.15} = \frac{0.1}{0.25}$$

d)

To function $f(x)$ is concave following equality must be satisfied where λ is between [0,1]

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$$

For $f(x) = |x|e^x$ equality can be written as follows

$$|\lambda x_1 + (1-\lambda)x_2|e^{\lambda x_1 + (1-\lambda)x_2} \leq \lambda|x_1|e^{x_1} + (1-\lambda)|x_2|e^{x_2}$$

Simplify equality as follows since exponentials are always positive

$$|\lambda x_1 + (1-\lambda)x_2| \leq \lambda|x_1| + (1-\lambda)|x_2|$$

From triangle inequality above function is satisfied. Hence, $f(x) = |x|e^x$ is a concave function.

SOLUTION FOR QUESTION 2

a)

The covariance matrix is always

b)

If And B are symmetric matrices and they have same eigenvectors but different eigenvalues, following matrices eigenvectors and eigenvalues can be calculated as follows. But note that resultant matrices does not have 0 as symmetric values.

i)

$$C = A + B$$

$$(A + B)u = \lambda v$$

$$\alpha u + \beta u = (\alpha + \beta)u = \lambda v$$

$$\lambda = (\alpha + \beta)$$

$$v = u$$

Resultant matrix C has eigenvalues that summation of eigenvalues A and B, and has same eigenvectors with others.

ii)

$$D = A - B$$

$$(A - B)u = \lambda v$$

$$\alpha u - \beta u = (\alpha - \beta)u = \lambda v$$

$$\lambda = (\alpha - \beta)$$

$$v = u$$

Resultant matrix D has eigenvalues that subtraction of eigenvalues A and B, and has same eigenvectors with others.

iii)

$$E = AB$$

$$(AB)u = \lambda v$$

$$\alpha u \beta u = (\alpha\beta)u = \lambda v$$

$$\lambda = (\alpha\beta)$$

$$v = u$$

Resultant matrix E has eigenvalues that multiplication of eigenvalues A and B, and has same eigenvectors with others.

iv) Taking an inverse of a symmetric matrix means their eigenvectors are same but eigenvalues are 1/eigenvalues

$$F = A^{-1}B$$

$$(A^{-1}B)u = \lambda v$$

$$\frac{1}{\alpha}u\beta u = \left(\frac{1}{\alpha}\beta\right)u = \lambda v$$

$$\lambda = \left(\frac{1}{\alpha}\beta\right)$$

$$v = u$$

Resultant matrix E has eigenvalues that multiplication of eigenvalues 1/A and B, and has same eigenvectors with others.

d)

Given equation $Ax = b$ can be solve as follows

$$x = A^{-1}Ab$$

i) if matrix A is non invertible, there is no solution.

ii) if A is invertible but columns of matrix A are linearly independent (which means one column of A can't be write as a multiplication of other columns) there is one unique solution

iii) if columns of matrix A are not linearly independent there is a multiple solution

SOLUTION FOR QUESTION 3

a)

From the Properties of gradient of Linear and Quadratic functions

$$f(x) = c^T x$$

$$\nabla f(x) = c$$

$$\nabla c = 0$$

$$\nabla_{x^2}{}^2 f(x) = 0$$

b)

Similarly since $M^T M$ is symmetric matrix. We can write in the following quadratic form where D denotes another symmetric matrix.

$$f(x) = M^T Mx = x^T M^T Mx = x^T Dx$$

$$\text{Where} D = M^T M$$

$$\nabla f(x) = (D^T + D)x$$

$$\nabla (D^T + D)x = D^T + D$$

$$\nabla_{x^2}{}^2 f(x) = D^T + D = MM^T + M^T M$$

İf Mt = M then following Equation will be obtained

$$\nabla_{x^2}{}^2 f(x) = D^T + D = MM^T + M^T M = 2M^T M$$

SOLUTION FOR QUESTION 4

a)

Second derivative of $H(p) = -p\log(p) - (1-p)\log(1-p)$ can be computed as follows

$$H(p) = -plog(p) - (1-p)\log(1-p)$$

$$H'(p) = \log(1-p) - \log(p)$$

$$H''(p) = \frac{1}{(p-1)p}$$

Since p is between [0-1] then following condition will always be satisfied

$$H''(p) < 0$$

b)

To find the value of maximizes the H(p), the value of p that set first derivative can be calculated

$$H'(p) = \log(1-p) - \log(p) = 0$$

$$H'(p) = \log\left(\frac{1-p}{p}\right) = 0$$

$$\frac{1-p}{p} = 1$$

$$p = 0.5$$

SOLUTION FOR QUESTION 5

a)

Conditional distribution of $y|\ x, w$ is as follows

$$P(y \mid x, w) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(y-x^T w)^2}{2\sigma^2}}$$

b)

Since they are independent, distribution of $y_1, y_2, \dots, y_n | x_1, x_2, \dots, x_n, w$ can be written as follows

$$P(y_1, y_2, \dots, y_n | x_1, x_2, \dots, x_n, w) = P(y_1|x_1, w)P(y_2|x_2, w) \dots P(y_n|x_n, w)$$

$$P(y_1, y_2, \dots, y_n | x_1, x_2, \dots, x_n, w) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(y_1-{x_1}^T w)^2}{2\sigma^2}}\right) \dots \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(y_n-{x_n}^T w)^2}{2\sigma^2}}\right)$$

c)

Log-likelihood function can be written as follows

$$L(D) = \prod_n^N P(y_n|x_n, w)$$

$$\ln\big(L(D)\big) = ln\left(\prod_n^N P(y_n|x_n, w)\right) = \sum_n^N \ln\big(P(y_n|x_n, w)\big)$$

$$= \sum_n^N \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(y_n-{x_n}^T w)^2}{2\sigma^2}}\right) = \sum_n^N \left(\ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) + \ln\left(e^{\frac{-(y_n-{x_n}^T w)^2}{2\sigma^2}}\right)\right)$$

$$= \sum_n^N \left(\ln\big((2\pi\sigma^2)^{-1/2}\big) - \frac{(y_n - {x_n}^T w)^2}{2\sigma^2}\ln(e)\right)$$

$$= \sum_n^N \left(\frac{-1}{2}\ln\big((2\pi)\big) - \frac{1}{2}\ln\big((\sigma^2)\big) - \frac{(y_n - {x_n}^T w)^2}{2\sigma^2}\right)$$

$$= \frac{-N}{2}\ln\big((2\pi)\big) - \frac{N}{2}\ln\big((\sigma^2)\big) - \sum_n^N \left(\frac{(y_n - {x_n}^T w)^2}{2\sigma^2}\right)$$

d)

To find the parameter w which maximizes log-likelihood function when d is 1, derivative with respect to w can be taken and set to zero as follows

$$\frac{d}{dw}\ln\big(L(D)\big) = \frac{d}{dw}\left(\frac{-N}{2}\ln\big((2\pi)\big) - \frac{N}{2}\ln\big((\sigma^2)\big) - \sum_{n}^{N}\left(\frac{(y_n - {x_n}^T w)^2}{2\sigma^2}\right)\right)$$

$$= 0 - 0 - \sum_{n}^{N}\frac{2x_n}{2\sigma^2}(y_n - x_n w)$$

From function above, parameter w can be calculated as follows

$$w = \sum_{n}^{N}\frac{x_n y_n}{{x_n}^2}$$

e)

Generalizing of equation parameter w as follows

$$\frac{d}{dw}\ln\big(L(D)\big) = \frac{d}{dw}\left(\frac{-N}{2}\ln\big((2\pi)\big) - \frac{N}{2}\ln\big((\sigma^2)\big) - \sum_{n}^{N}\left(\frac{(y_n - {x_n}^T w)^2}{2\sigma^2}\right)\right)$$

$$= 0 - 0 - \frac{d}{dw}\left(\sum_{n}^{N}\left(\frac{(y_n - {x_n}^T w)^2}{2\sigma^2}\right)\right) = \frac{1}{2\sigma^2}\sum_{n}^{N} 2(y_n - {x_n}^T w)\,\nabla_w\,(y_n - {x_n}^T w)$$

$$\sum_{n}^{N}[(y_n - {x_n}^T w)x_n]$$

SOLUTION FOR QUESTION 6